DATA
DRIVEN
ENVIROLAB

Data scientist/Research specialist (pre-doctoral fellow)

Technical Assignment

Task Description

You provide support to an organization in curating their database and integrating the external databases into their database well as managing their products

Assignment A Instructions

- Update the NetZeroTracker database with the 2026 Forbes2000 database. Integration of the companies should happen in a single script using R or Python with the columns for **sales and profit** appended to the original NetZeroTracker data.
- Identify how many of the entities in 2026 Forbes2000 list are not currently included in the NetZeroTracker Database, along with a brief explanation of possible matching errors.

Assignment B Instructions

- Use the Climactor v2 data to construct **slides within 6 pages** of the most relevant data from the dataset. Include at least one spatial visualization (be creative)
- Indicate how did you decide on which features to highlight and what talk about data quality, how to manage the data, and potential updates.
- What are the most relevant aspects that need to be monitored for proper functioning and maintenance.

Demonstration and Explanation

The use of AI-assisted tools (including ChatGPT, GitHub Copilot, Claude, or similar tools) is **not permitted for Assignment A**. All code, data processing, entity matching, and analytical decisions must be completed independently.

You may use AI coding tools for Assignment B, but we expect the logic of the workflow to be defined by you and not by the AI tool. Be prepared to explain the method, error handling, and limitations. We are interested in your understanding of the implementation, not only the final output.

Submission Requirements

Provide your code and instructions to run it, your answers to the questions, and a short note describing your approach and limitations. Please do not include API keys. The deadline for the submission is **10/18 11:59pm**

GitHub Submission Instructions

Please submit your assignment through GitHub using the following steps:

1. Fork the assignment repository: [https://github.com/datadrivenenvirolab/research_specialist_test.git].
2. Clone your fork to your local computer.
3. Create a folder named `Firstname_Lastname_DDL_test` in the repository.
4. Place all your code, datasets, presentation files, and README inside this folder.
5. Commit and push your changes to your fork.
6. Open a pull request to the original repository, titled `Technical Assignment: Firstname Lastname`.

7. Include a brief summary of your submission in the pull request description.

Please do not modify or delete the original assignment files or other folders.

Your submission is complete once you have opened the pull request. You do not need to merge it.

We look forward to reviewing your submission.